

Client Application Program Manual

Digiflash – PETC Data Submission Client

DOTr IT Provider Accreditation – Deliverable #1

Document Control

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Prepared for	Department of Transportation (DOTr) / Land Transportation Office (LTO)
Audience	Trained PETC operators (encoders and supervisors)
Language	English

Revision History

Version	Date	Author	Summary
0.1	2026-06-02	C. Silerio	Initial draft for DOTr accreditation submission.
0.2	2026-06-03	C. Silerio	Aligned wizard chapter and printing chapter with the implemented code: corrected Step-1 vehicle fields incl. classification, Step-2 owner fields, Step-3 engine flags (turbo / aspiration / condition), Step-4 technician fields, Step-5 single mandatory FRONT photo. Added WAITING_FOR_LTMS outcome + matching History/troubleshooting entry. Split the CEC printing chapter into the A4 PDF (formal) and thermal-receipt (hand-off) outputs. Clarified that LTMS / IRDS lookups are cloud-proxied.

1. Introduction

1.1 Purpose

This manual describes the day-to-day operation of the **Digiflash PETC Data Submission Client** at an accredited Private Emission Testing Center. It is written for the operators who will actually run the application — encoders and supervisors — and is structured so a new operator can complete a full emission test from login to printed CEC by following the procedures in order.

1.2 Product Overview

Digiflash is a Windows desktop application installed at each accredited PETC. It:

1. Captures emission readings from the connected analyzer (Fofen petrol gas analyzer or Fofen diesel opacimeter at the pilot center).
2. Photographs the vehicle through a USB webcam.
3. Looks up vehicle and owner data from LTMS / IRDS **through the Digiflash cloud** (the desktop never contacts LTMS or Stradcom directly).
4. Combines all of the above into a single emission test record.
5. Submits the completed record to LTMS and IRDS through the Digiflash cloud service.
6. Produces a Certificate of Emission Compliance (CEC) in two forms once LTMS returns the certificate key: an **A4 PDF** (two halves per sheet — Customer copy + Center copy, the formal record kept on file at the center) and an **80 mm thermal receipt** (a printed summary the operator hands to the vehicle owner on the spot).

The desktop application is the source of truth at the center. A center can continue capturing tests during a temporary internet outage; queued tests will be uploaded automatically when connectivity is restored.

1.3 Roles

Role	Allowed actions
Encoder	Log in, run tests, capture photos, complete the LTMS upload wizard, print CECs, view own test history.
Supervisor / Admin	All encoder actions plus: manage operator accounts at the center, configure analyzer / camera / printer, review and re-print past CECs, view the daily summary report.
Platform Super Admin	Operator-cloud only; not available inside the desktop application.

2. System Requirements

2.1 Workstation

Item	Minimum specification
Operating System	Windows 10 (64-bit, version 21H2 or newer) or Windows 11
CPU	Intel Core i3 (10th gen) or AMD Ryzen 3 (3000 series)

RAM	8 GB
Storage	256 GB SSD with at least 50 GB free
USB ports	3 free USB-A ports (analyzer adapter, webcam, printer)
Display	1366 × 768 minimum (1920 × 1080 recommended)
Internet	Stable broadband, minimum 5 Mbps down / 2 Mbps up
Power	UPS with at least 30 minutes of runtime at idle PC + monitor + printer

2.2 Peripherals at the Testing Bay

Item	Specification
Emission analyzer	Fofen petrol gas analyzer (gas vehicles) and Fofen diesel opacimeter (diesel vehicles), connected via USB-to-Serial adapter
Webcam	USB 2.0 webcam, 1080p (2 MP), fixed focus, ≥60° field of view, ≤1 lux low-light. See 02–setup-and-network-layout.md §4 for the functional-compliance basis.
Receipt printer	80 mm thermal receipt printer, USB, ESC/POS-compatible
Network	Wired Ethernet preferred; Wi-Fi acceptable with a strong signal

3. Installation

Installation is performed by the Digiflash field engineer during initial PETC commissioning. Operators do not normally need to install the software themselves; this section is included for completeness.

3.1 Steps

1. Sign in to Windows with an administrator account.
2. Run the installer `Digiflash-PETC-Setup-<version>.exe` provided by Digiflash.
3. Accept the license terms and proceed through the installer.
4. When prompted, enter the **Center ID** and **Center API Key** issued by Digiflash.
5. Choose the install location (default `C:\Program Files\Digiflash`) and complete the installer.
6. The installer creates a desktop shortcut named **Digiflash PETC**.
7. Reboot if prompted.

3.2 First Launch

On first launch, the application performs a self-check and may prompt for:

- The COM port and baud rate of the Fofen analyzer.
- The USB device of the webcam.
- The USB device of the thermal printer.

These settings are saved and do not need to be re-entered.

Note — installer screenshots. *The installer welcome screen and the first-launch hardware self-check are produced by a signed, packaged Windows build (`Digiflash-PETC-Setup-<version>.exe`). These screens do not exist in the reference / development environment, which runs the application directly from source. The corresponding screenshots will be inserted here once a signed packaged build is produced.*

4. Logging In

4.1 Operator Login

1. Double-click the **Digiflash PETC** shortcut to launch the application.
2. The login screen appears.
3. Enter your registered **email** and **password**.
4. Click **Sign In**.

If your credentials are correct, you are taken to the home screen. If they are incorrect, an inline error message is shown; after five consecutive failures the account is temporarily locked for 15 minutes.

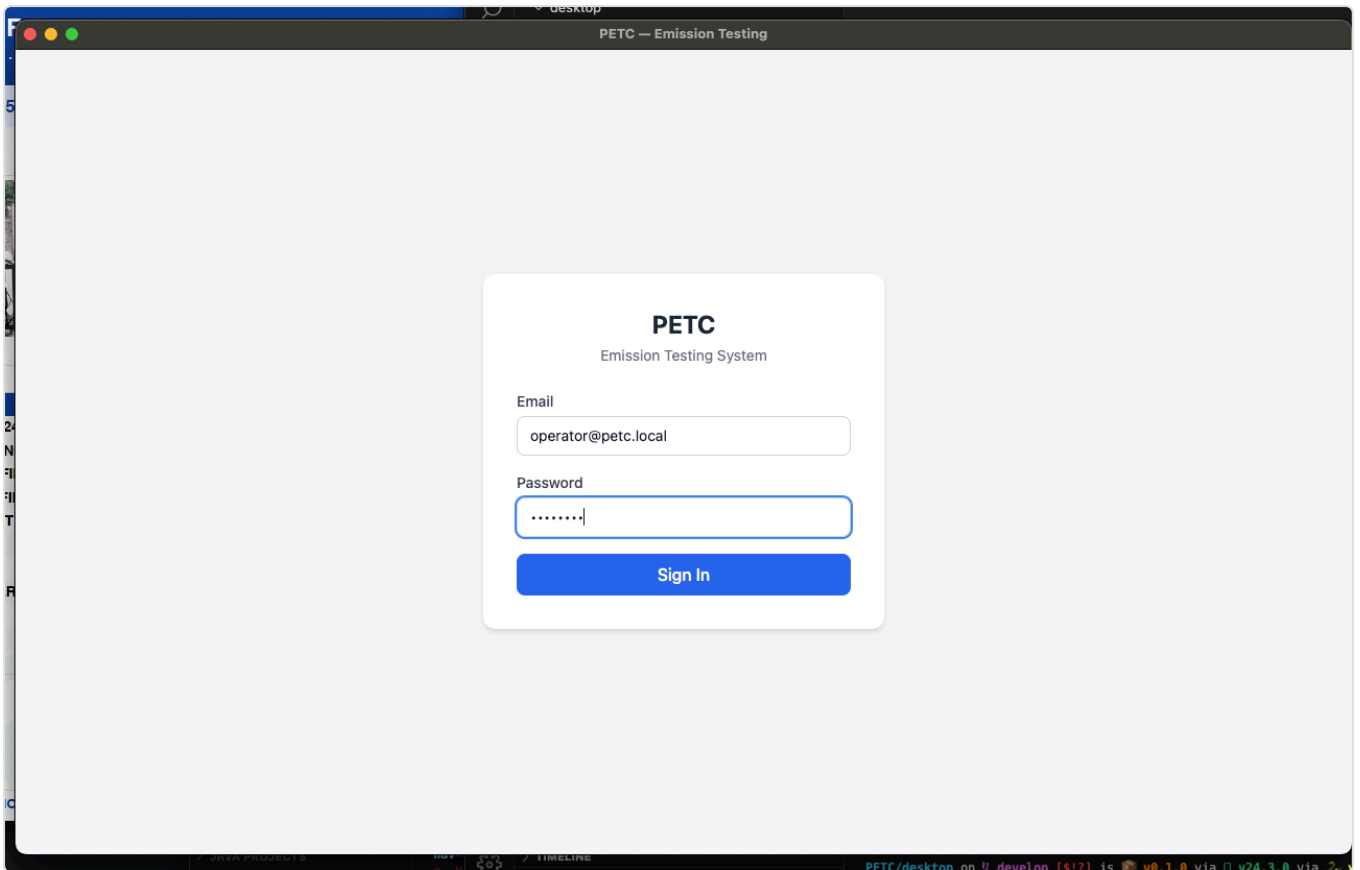


Figure 4.1 — Login screen with empty fields.

4.2 Forgotten Password

Ask the center supervisor to reset your password from the **Settings** → **Operators** screen. Password resets are not self-service on the desktop application.

4.3 Logging Out

Click the operator name in the top-right corner of any screen and choose **Sign Out**. The application returns to the login screen and the local session token is discarded.

5. Home Screen Overview

After login, the home screen shows:

- The current operator and role in the top-right.
- The center name in the top-left.
- A primary action button: **Run Test**.
- A secondary navigation strip with: **Run Test**, **LTMS Upload**, **History**, **Analytics**, **Settings**.

- A status bar at the bottom showing analyzer connection status, printer status, and cloud sync status.

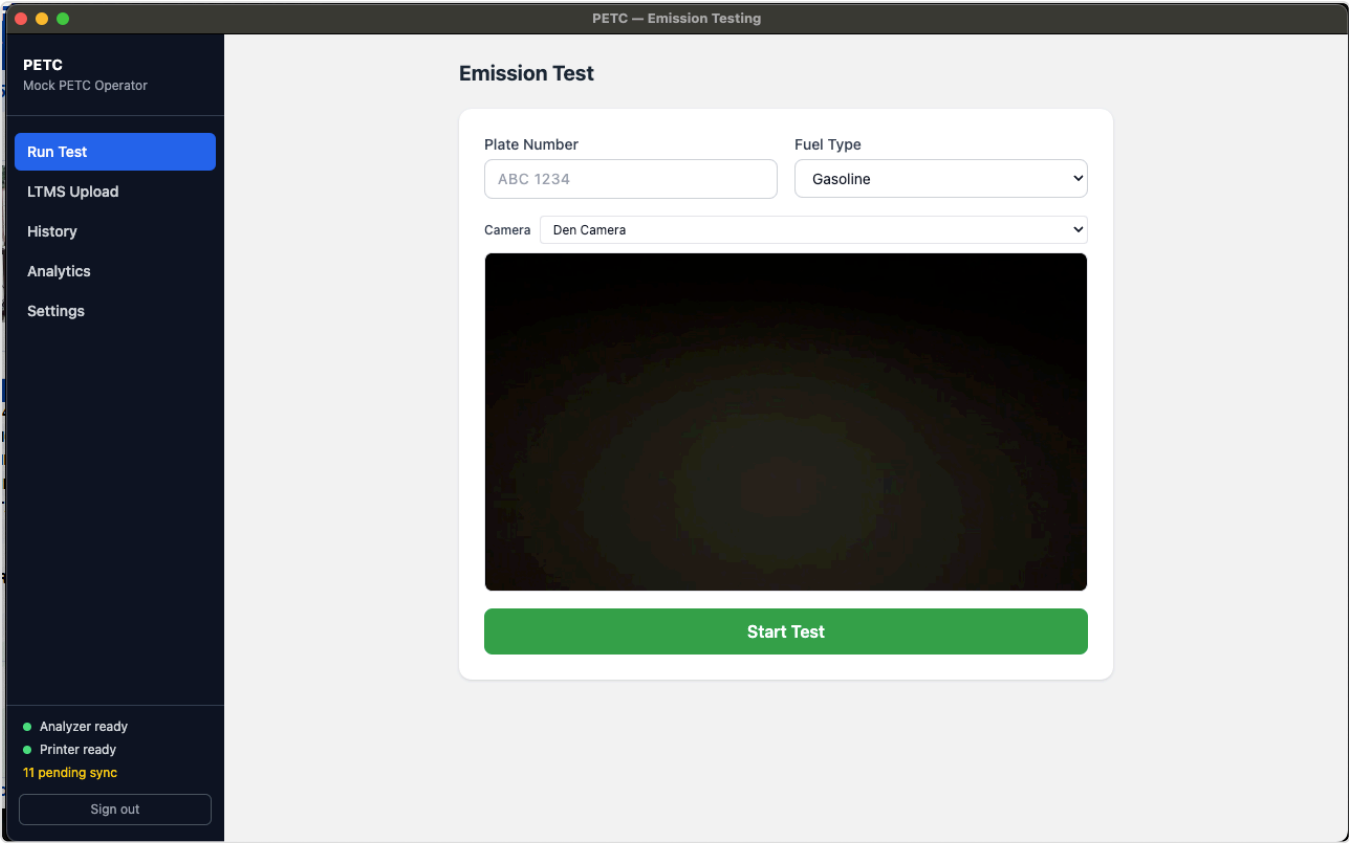


Figure 5 — Home screen with status bar at the bottom.

5.1 Status Bar Indicators

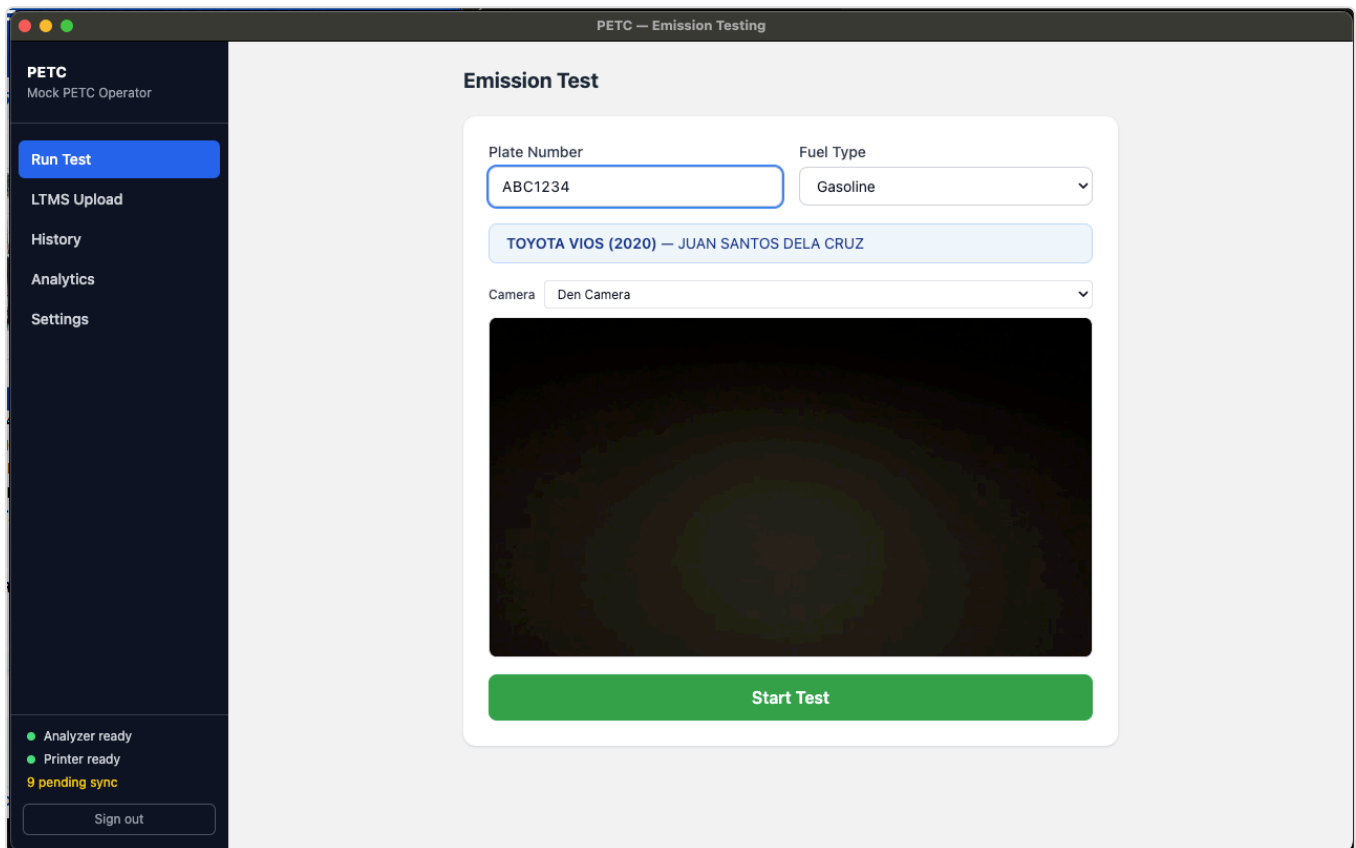
Indicator	Healthy state	What it means if unhealthy
Analyzer	Green: connected	Red: analyzer COM port not opened; check the cable.
Camera	Green: detected	Red: webcam not detected; re-plug USB.
Printer	Green: ready	Red: printer offline, no paper, or driver issue.
Cloud	Green: synced	Yellow: backlog of tests waiting to upload; will auto-retry. Red: cloud unreachable; tests are still saved locally.

6. Running an Emission Test

This is the core daily workflow.

6.1 Starting a Test

1. From the home screen, click **Run Test**.
2. Enter the vehicle's plate number in the **Plate Number** field.
3. Click **Look Up**. Digiflash queries LTMS / IRDS through the cloud and pre-fills the vehicle and owner details.
 - If the plate is not found, choose **Continue without lookup** to enter details manually.
4. Choose the fuel type:
 - **Gasoline** routes the test to the Fofen petrol gas analyzer.
 - **Diesel** routes the test to the Fofen diesel opacimeter.
5. Click **Start Test**.



The screenshot displays the 'PETC — Emission Testing' application interface. On the left is a dark sidebar with the title 'PETC Mock PETC Operator' and a menu containing 'Run Test' (highlighted in blue), 'LTMS Upload', 'History', 'Analytics', and 'Settings'. At the bottom of the sidebar are status indicators: 'Analyzer ready' (green dot), 'Printer ready' (green dot), '9 pending sync' (yellow dot), and a 'Sign out' button. The main content area is titled 'Emission Test' and contains a form. The form has a 'Plate Number' input field with 'ABC1234' entered, a 'Fuel Type' dropdown menu set to 'Gasoline', and a 'Camera' dropdown menu set to 'Den Camera'. Below these fields is a light blue box displaying 'TOYOTA VIOS (2020) — JUAN SANTOS DELA CRUZ'. A large black rectangular area represents a video feed from the camera. At the bottom of the form is a prominent green button labeled 'Start Test'.

Figure 6.1 — Run Test screen, plate lookup completed.

6.2 Capturing Analyzer Readings

After clicking **Start Test**, Digiflash opens the connection to the configured analyzer and begins receiving readings.

For **gasoline** tests, the following fields are captured: CO, HC, CO₂, O₂, λ (lambda), engine RPM, oil temperature, analyzer serial number, and the analyzer's pass/fail indication.

For **diesel** tests, the following fields are captured: opacity (%), k-value (m⁻¹), engine RPM, boost pressure, analyzer serial number, and the analyzer's pass/fail indication.

The screen updates live until the analyzer signals end-of-test. The final accepted reading is then displayed.

If the analyzer disconnects or returns no data within 60 seconds, an error banner appears with the suggested fix.

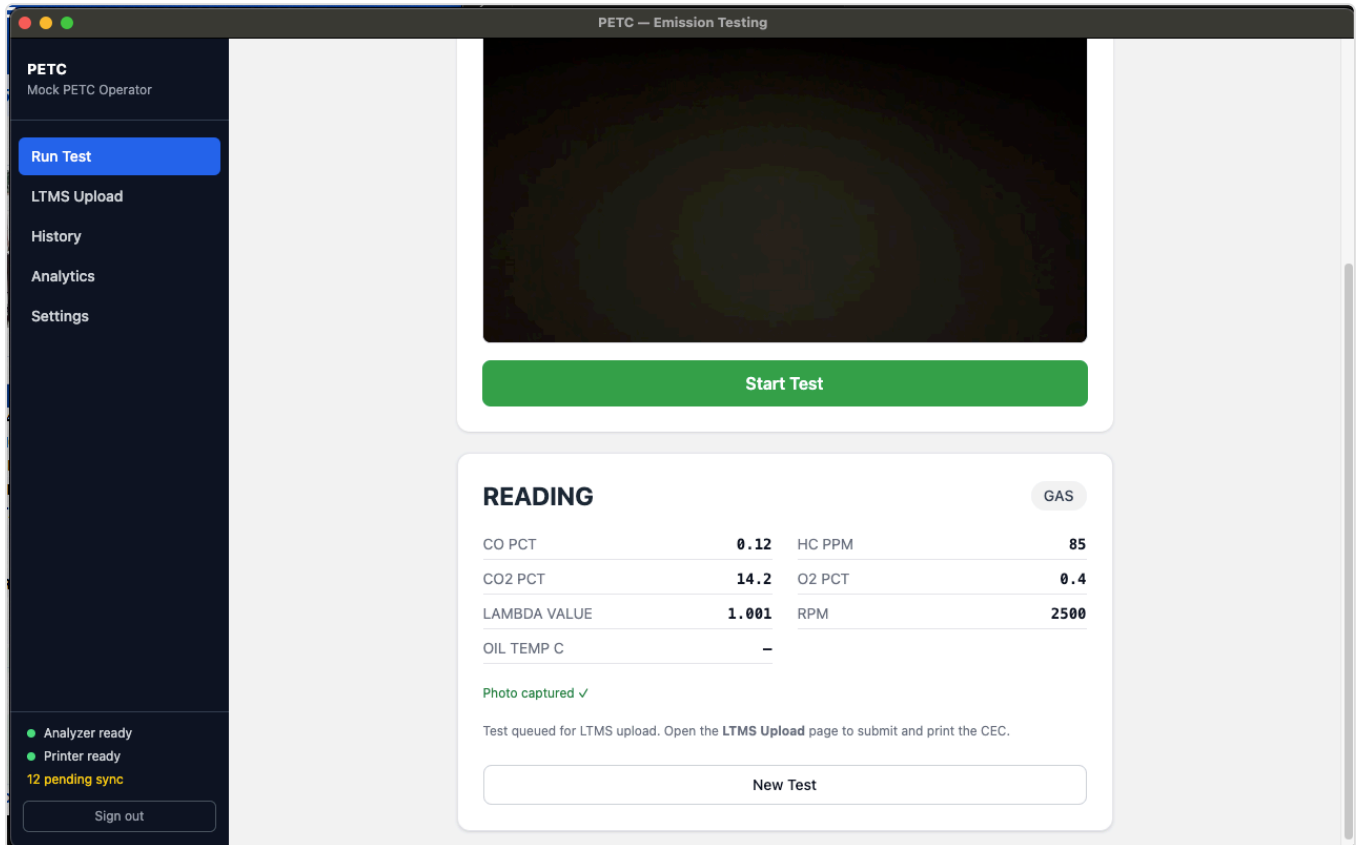


Figure 6.2 — Live analyzer readings during a diesel test.

6.3 Capturing Photos

After analyzer readings are captured:

1. Position the vehicle so the rear plate is visible in the webcam preview.
2. Click **Capture Photo**.
3. Repeat to capture at least:
 - One photo showing the vehicle and rear plate.
 - One photo showing the vehicle at the testing bay.

Each photo is stored locally and tagged with a SHA-256 hash, the test ID, the operator ID, and a timestamp.

Note — photo-capture screenshot. The photo-capture screen requires a connected webcam and a vehicle at the testing bay, which are only available at an operating PETC. This screen is captured on-site during inspection rather than in the reference / development environment, and the corresponding screenshot will be inserted here at that time.

6.4 Saving the Test

Click **Save Test**. The test is saved to the local database with status **Pending Upload**. You can now either:

- Continue to **LTMS Upload** to submit the test immediately, or
- Return to the home screen and submit later from the **LTMS Upload** queue.

7. LTMS Upload Wizard

The LTMS Upload wizard takes a saved test through a six-step review before submission. It is mandatory for every test.

7.1 Opening the Wizard

1. From the navigation strip, click **LTMS Upload**.
2. A list of pending tests is shown, newest first.
3. Click the row of the test you want to submit.

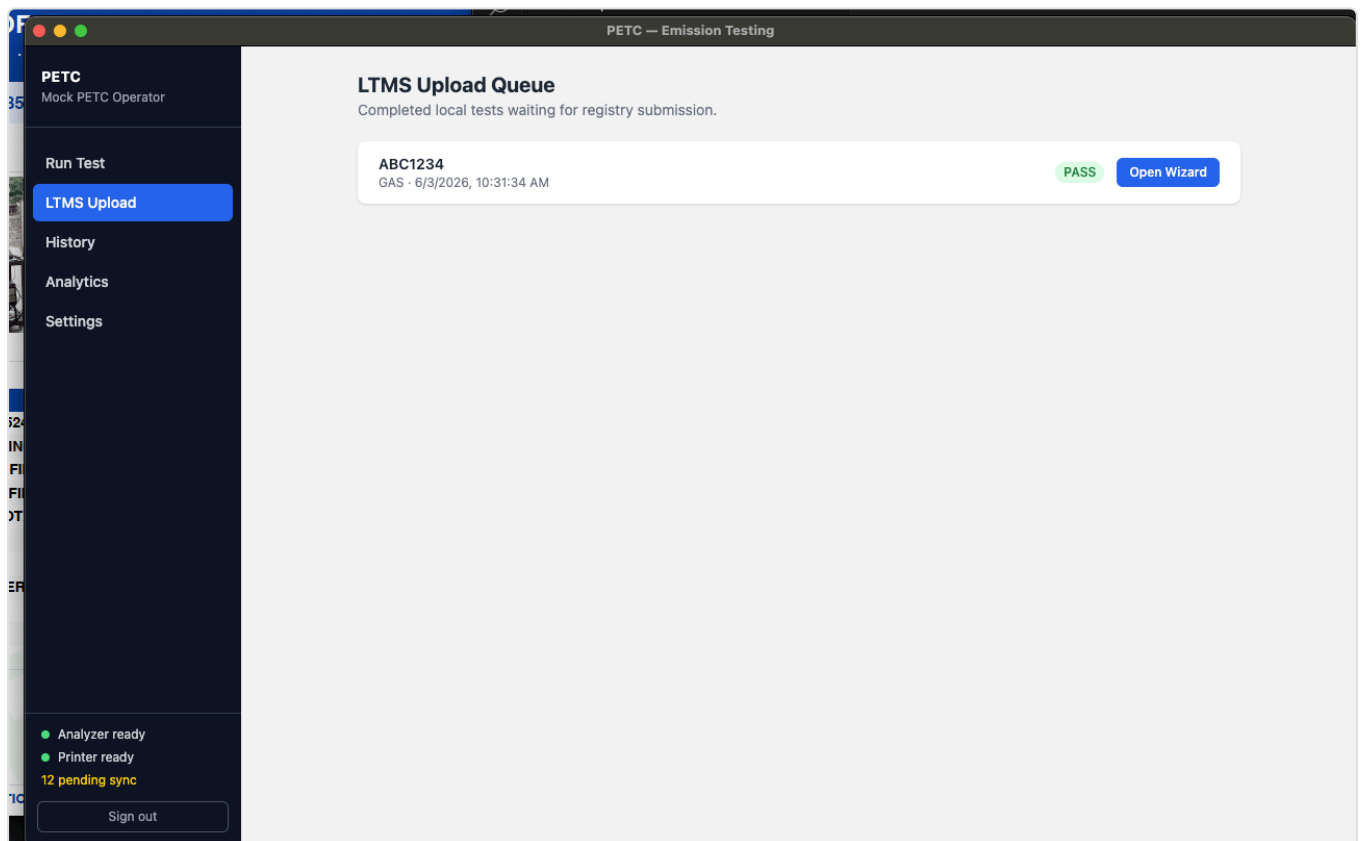


Figure 7.1 — Pending uploads list.

7.2 Step 1 – Vehicle Details

Review and correct (if needed) the following fields, most of which are pre-filled from the LTMS plate lookup. Fields the LTMS lookup populated are tagged **From LTMS**; fields you change are re-tagged **Edited**.

- Plate No, MV File No, Engine No, Chassis No
- OR Type, CR Date, CR No, District Office
- Make, Series, Vehicle Type, Year Model, Color
- Transmission (M/T or A/T)
- Fuel Type (GAS, DIESEL, or MOTORCYCLE)
- **Classification** (PRIVATE, PUBLIC, GOVERNMENT, or DIPLOMATIC) — this prints on the CEC.

Plate No, Make, Series, Engine No, and Chassis No are required before the wizard will let you proceed. Click **Next**.

The screenshot shows the PETC Emission Testing interface. The left sidebar contains navigation links: PETC, Mock PETC Operator, Run Test, LTMS Upload (highlighted), History, Analytics, and Settings. The main content area is titled 'LTMS Upload - ABC1234' and shows a progress bar with steps: 1. Vehicle (active), 2. Owner, 3. Results, 4. Technician, 5. Photos, and 6. Review. The 'STEP 1 - PLATE LOOKUP + VEHICLE DETAILS' form includes the following fields:

- Plate No: ABC1234
- MV No: From LTMS (MV-ABC1234)
- Engine No: From LTMS (ENG-ABC1234)
- Chassis No: From LTMS (CHS-ABC1234)
- OR Type: From LTMS (MVR)
- CR Date: From LTMS (01/18/2025)
- CR No: From LTMS (CR-ABC1234)
- District Office: From LTMS (1368 - PASAY CITY DISTRICT OFFICE)
- Make: From LTMS (TOYOTA)
- Series: From LTMS (VIOS)
- Vehicle Type: From LTMS (CAR)
- Year Model: From LTMS (2020)
- Color: From LTMS (SILVER)
- Transmission: M/T, A/T (selected)
- Fuel Type: GAS (selected), DIESEL, MOTORCYCLE
- Classification: PRIVATE (selected), PUBLIC, GOVERNMENT, DIPLOMATIC

A 'Next' button is located at the bottom right of the form. The bottom left of the sidebar shows status indicators: Analyzer ready, Printer ready, 12 pending sync, and a Sign out button.

Figure 7.2 — Wizard Step 1, vehicle details.

7.3 Step 2 – Owner Details

Choose the **Owner Type**:

- **INDIVIDUAL** — enter Last Name, First Name, Middle Name, Address, City.
- **ORGANIZATION** — enter Organization name, Address, City.

For individual owners, Last Name, First Name, Address, and City are required. For organisation owners, Organization, Address, and City are required. The wizard will not advance until the required fields are filled.

Click **Next**.

The screenshot shows the PETC Emission Testing interface. The top bar indicates 'PETC — Emission Testing'. The main header shows 'LTMS Upload - ABC1234' with a 'Back' link and a 'Lookup Plate' button. Below the header is a progress bar with six steps: 1. Vehicle (highlighted in green), 2. Owner (highlighted in blue), 3. Results, 4. Technician, 5. Photos, and 6. Review. The main content area is titled 'STEP 2 - VEHICLE OWNER'. It contains a form for owner details. The 'Owner Type' is set to 'INDIVIDUAL'. The 'Last Name' is 'DELA CRUZ', 'First Name' is 'JUAN', 'Middle' is 'SANTOS', and 'Address' is 'Roxas Boulevard'. The 'City' is 'Pasay City'. There are 'Back' and 'Next' buttons at the bottom of the form. On the left sidebar, there are links for 'Run Test', 'LTMS Upload' (highlighted), 'History', 'Analytics', and 'Settings'. At the bottom of the sidebar, there are status indicators: 'Analyzer ready', 'Printer ready', and '12 pending sync', along with a 'Sign out' button.

Figure 7.3 — Wizard Step 2, owner details.

7.4 Step 3 – Engine Flags and Readings

Review the captured analyzer readings shown on this step. The verdict (PASS / FAIL) is computed automatically from the readings against the configured emission limits and is shown in a green or red banner with the supporting reasons.

Set the three engine flags shown above the readings:

- **Turbo** — TURBO or NON_TURBO.
- **Aspiration** — ELEVATION or N_ASPIRATED.
- **Condition** — REBUILT or CONVENTIONAL.

These flags appear on the LTMS submission and influence how LTMS interprets the readings. The verdict updates live when you change a flag.

Click **Next**.

The screenshot shows the PETC Emission Testing interface. The sidebar on the left contains the following items: PETC Mock PETC Operator, Run Test, LTMS Upload (highlighted), History, Analytics, and Settings. At the bottom of the sidebar, there are status indicators: Analyzer ready (green dot), Printer ready (green dot), 12 pending sync (yellow text), and a Sign out button. The main content area is titled 'LTMS Upload - ABC1234' with a 'Back' link and a 'Lookup Plate' button. A progress bar shows six steps: 1. Vehicle, 2. Owner, 3. Results (active), 4. Technician, 5. Photos, and 6. Review. The 'STEP 3 - ENGINE FLAGS + EMISSION TEST RESULTS' section contains several fields: Turbo (TURBO, NON.TURBO), Aspiration (ELEVATION, N.ASPIRATED), Condition (REBUILT, CONVENTIONAL), CO PCT (0.12), HC PPM (85), CO2 PCT (14.2), O2 PCT (0.4), LAMBDA VALUE (1.001), RPM (2500), and OIL TEMP C (N/A). A green box displays 'GAS verdict: PASS'. At the bottom of the main content area are 'Back' and 'Next' buttons.

STEP 3 - ENGINE FLAGS + EMISSION TEST RESULTS					
Turbo		Aspiration		Condition	
☐ TURBO ☒ NON.TURBO		☐ ELEVATION ☒ N.ASPIRATED		☐ REBUILT ☒ CONVENTIONAL	
CO PCT	HC PPM	CO2 PCT	O2 PCT		
0.12	85	14.2	0.4		
LAMBDA VALUE	RPM	OIL TEMP C			
1.001	2500	N/A			
GAS verdict: PASS					

Figure 7.4 — Wizard Step 3, engine flags and readings.

7.5 Step 4 – Technician and Certification

Three fields are shown, pre-filled from your operator profile:

- **Technician Name** — the certifying technician's name as it should appear on the CEC.
- **TESDA Cert. No** — TESDA certification number.
- **Certification No** — the PETC technician certification number issued under DOTr / LTO accreditation.

All three are required. Correct any errors before continuing. Click **Next**.

PETC — Emission Testing

PETC
Mock PETC Operator

Run Test

LTMS Upload

History

Analytics

Settings

Analyzer ready

Printer ready

12 pending sync

Sign out

Back

LTMS Upload - ABC1234
From LTMS cache (6/3/2026, 10:21:37 AM)

Lookup Plate

1. Vehicle 2. Owner 3. Results 4. Technician 5. Photos 6. Review

STEP 4 - TECHNICIAN / CERTIFICATION

Technician Name

Mock PETC Operator

TESDA Cert. No

TESDA-MOCK-001

Certification No

PETC-CERT-MOCK

Back

Next

Figure 7.5 — Wizard Step 4, technician and certification.

7.6 Step 5 – Photos

A single **Vehicle photo** (type **FRONT**) is required. If the photo has not yet been captured, position the vehicle so the plate is visible in the webcam preview on the left of the screen and click **Take photo**. To replace an existing photo, click **Retake photo**. The mandatory photo is shown as a thumbnail on the right when captured.

Click **Next** once at least one Vehicle photo is attached. (A second supplementary photo can be captured if the operator wishes — the LTMS submission and the printed CEC support a second photo slot.)

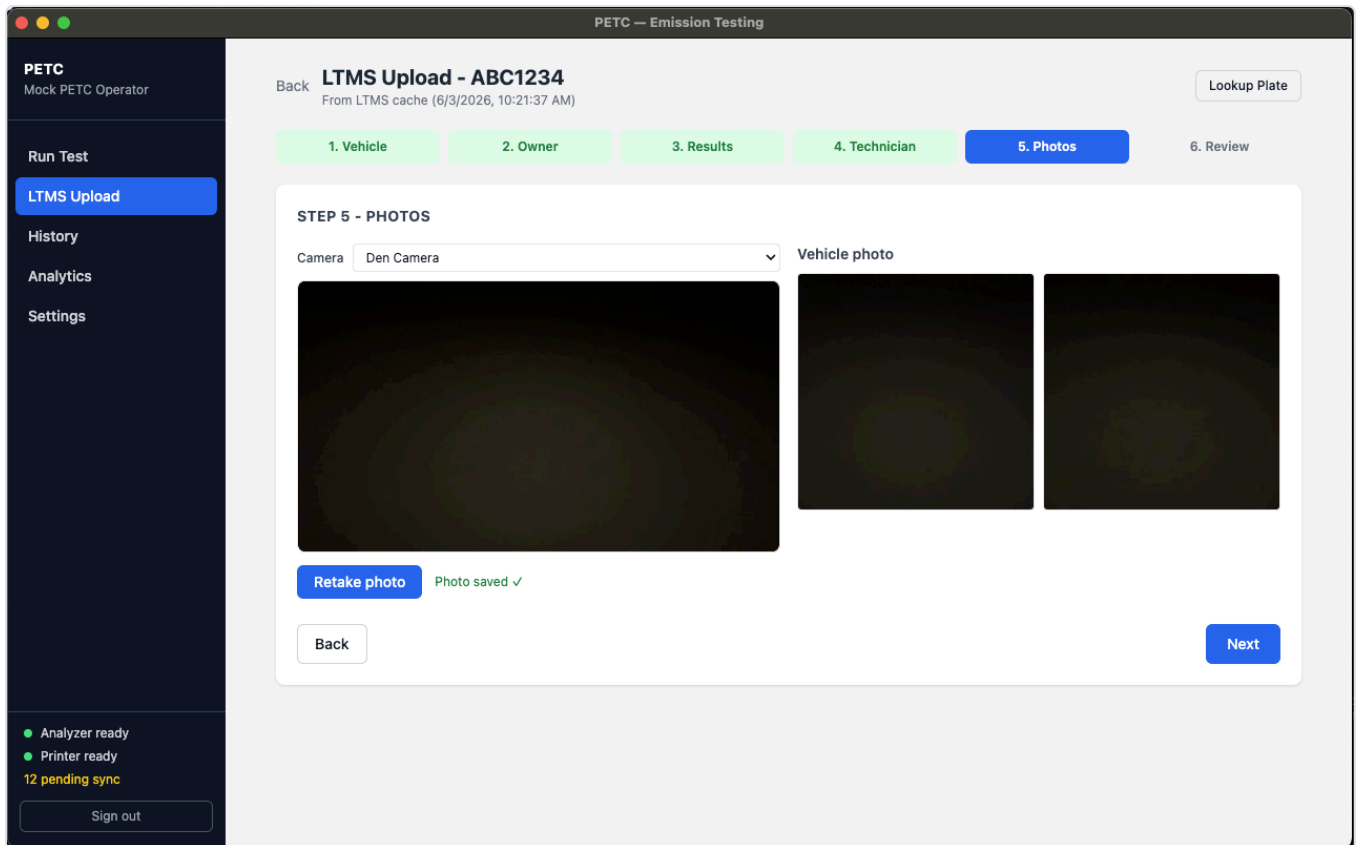


Figure 7.6 — Wizard Step 5, photos.

7.7 Step 6 – Review and Submit

The final screen shows a read-only summary of all data going to LTMS and IRDS. Verify carefully. Click **Submit to LTMS**.

The wizard shows a progress indicator while the Digiflash cloud submits the record to LTMS and IRDS on behalf of the center.

PETC
Mock PETC Operator

Run Test

LTMS Upload

History

Analytics

Settings

● Analyzer ready
● Printer ready
13 pending sync

Sign out

PETC — Emission Testing

Back **LTMS Upload - ABC1234**
From LTMS cache (6/3/2026, 10:21:37 AM) [Lookup Plate](#)

1. Vehicle 2. Owner 3. Results 4. Technician 5. Photos 6. Review

STEP 6 - REVIEW & SUBMIT

VEHICLE
Plate ABC1234
Vehicle 2020 TOYOTA VIOS
Fuel GAS
Engine / Chassis ENG-ABC1234 / CHS-ABC1234

OWNER
Type INDIVIDUAL
Name JUAN SANTOS DELA CRUZ
Address Roxas Boulevard, Pasay City

VERDICT
Result PASS
Reason Within configured mock limits

PHOTOS
Attached 2

Back [Submit to LTMS](#)

Figure 7.7 — Wizard Step 6, review and submit.

7.8 Outcomes

The wizard waits up to 60 seconds for LTMS to return a verdict. Possible outcomes:

Outcome	What you will see	What to do
Accepted	A green confirmation with the CEC number (CERT-...), the LTMS OR No., and the option Print CEC . The A4 CEC PDF is rendered in a preview pane.	Click Print CEC x2 to send the A4 PDF to the printer for the customer + center copies, and use /print/receipt at the bay (or the History row's print action) to hand the owner a thermal-receipt summary.
Rejected	A red banner showing the rejection reason returned by LTMS / IRDS. No CEC is issued.	Correct the noted issue, then re-submit from the LTMS Upload queue.
Awaiting LTMS (WAITING_FOR_LTMS)	A blue banner: "Queued — awaiting LTMS response." The test has been received by the Digiflash cloud but LTMS has not yet returned a verdict within the 60-second window.	Click Go to History . The Digiflash desktop will check LTMS every 30 seconds in the background; when LTMS responds, the History row updates automatically — the Print CEC button appears for an Accepted result, or the rejection reason appears for a Rejected one.

Queued (offline)	A yellow banner: "Saved. Will upload when connection is restored."	Continue working; the upload will retry automatically when connectivity returns.
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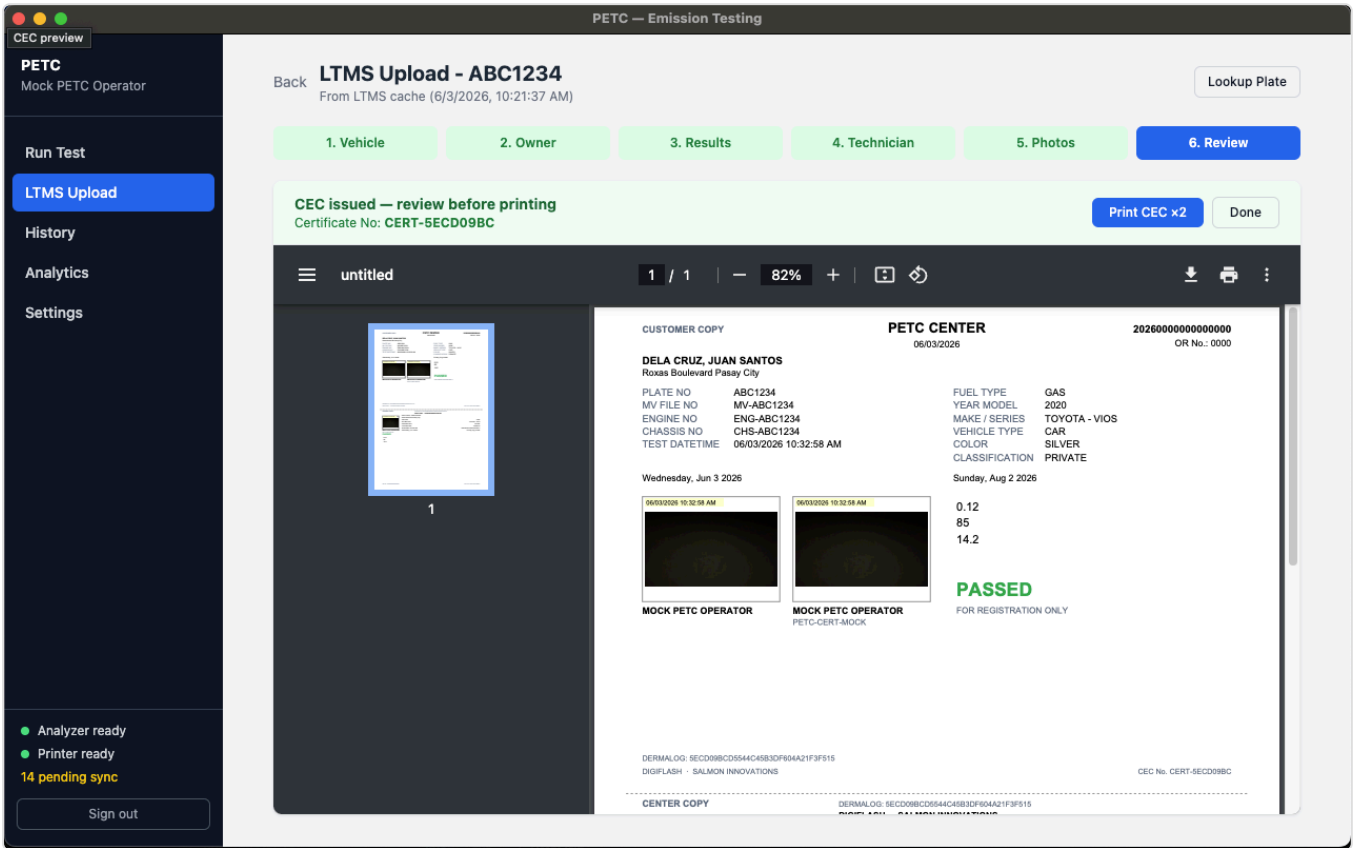


Figure 7.8 — Accepted result with print button and A4 CEC preview.

8. Printing the CEC

Digiflash produces two outputs for every Accepted submission. Both are issued only after LTMS returns the certificate number (`CERT-...`), OR number, and DERMALOG token; nothing is printed before LTMS accepts the submission.

8.1 A4 CEC PDF (formal record)

The A4 PDF is the formal Certificate of Emission Compliance described in deliverable #5 (`05-cec-samples/`). One A4 sheet carries two halves:

1. **Customer Copy** (top half) — given to the vehicle owner for presentation at LTO during registration renewal.
2. **Center Copy** (bottom half) — retained by the PETC for the regulatory audit trail.

Both halves carry the same CEC No., OR No., DERMALOG cryptographic token, validity window, vehicle and owner identifiers, technician identification, emission readings, and verdict.

To print: click **Print CEC x2** on the wizard result screen (or on the History row for an Accepted submission). The PDF can also be previewed in the application before printing via the embedded viewer on the result screen, and re-opened from History.

8.2 Thermal Receipt (operator hand-off)

A short thermal-receipt summary is produced on the 80 mm thermal printer for the operator to hand the owner immediately. It contains the plate, vehicle make/model/year, fuel type, verdict, technician name, center name, CEC number, and timestamp. The thermal receipt is a **convenience artifact** — the A4 PDF is the formal CEC.

The thermal receipt is printed via the **Print Receipt** action available on the Run-Test result screen and at the History row.

8.3 Re-printing a CEC

1. From the navigation strip, click **History**.
2. Find the test by plate number, date, or CEC number. Rows whose state is **ACCEPTED** show a **Print CEC** button on the right.
3. Click **Print CEC** to re-print the A4 PDF, or use the receipt action to re-print the thermal slip.

Re-prints are logged in the audit trail with the operator ID and timestamp.

The screenshot displays the PETC Emission Testing application interface. On the left is a dark navigation sidebar with the following items: PETC (Mock PETC Operator), Run Test, LTMS Upload, History (highlighted in blue), Analytics, and Settings. At the bottom of the sidebar, there are status indicators: a green dot for 'Analyzer ready', a green dot for 'Printer ready', and a yellow dot for '14 pending sync', along with a 'Sign out' button. The main content area is titled 'Test History' and contains a list of five test entries. Each entry shows a plate number, fuel type, date and time, a 'PASS' status, an 'ACCEPTED' status, and a 'Print CEC' button. The entries are: ABC1234 (GAS, 6/3/2026, 10:31:34 AM, Cert: CERT-5ECD09BC), DSL123 (DIESEL, 6/3/2026, 10:28:51 AM, Cert: CERT-79A60E7F), ABC1234 (GAS, 6/3/2026, 10:27:53 AM, Cert: CERT-A6363754), DSL1234 (DIESEL, 6/3/2026, 10:23:17 AM, Cert: CERT-859C8010), and ABC1234 (GAS, 6/3/2026, 10:21:41 AM, Cert: CERT-9B77A080).

Plate	Fuel	Date/Time	Cert	Status	Action
ABC1234	GAS	6/3/2026, 10:31:34 AM	CERT-5ECD09BC	PASS	ACCEPTED Print CEC
DSL123	DIESEL	6/3/2026, 10:28:51 AM	CERT-79A60E7F	PASS	ACCEPTED Print CEC
ABC1234	GAS	6/3/2026, 10:27:53 AM	CERT-A6363754	PASS	ACCEPTED Print CEC
DSL1234	DIESEL	6/3/2026, 10:23:17 AM	CERT-859C8010	PASS	ACCEPTED Print CEC
ABC1234	GAS	6/3/2026, 10:21:41 AM	CERT-9B77A080	PASS	ACCEPTED Print CEC

Figure 8.3 — History row with Print CEC button next to an ACCEPTED state badge.

9. History and Daily Summary

9.1 History

The **History** screen lists every test conducted at this center, with filters for date range, plate number, fuel type, and result. Encoders see only their own tests by default; supervisors see all tests at the center.

9.2 Daily Summary (Supervisor only)

Open **Analytics** → **Daily Summary**. The page shows:

- Total tests today, broken down by gas vs diesel.
- Pass / fail count.
- Number of LTMS submissions accepted, rejected, and still queued.
- Operator-level activity.

These numbers reset at midnight (PHT).

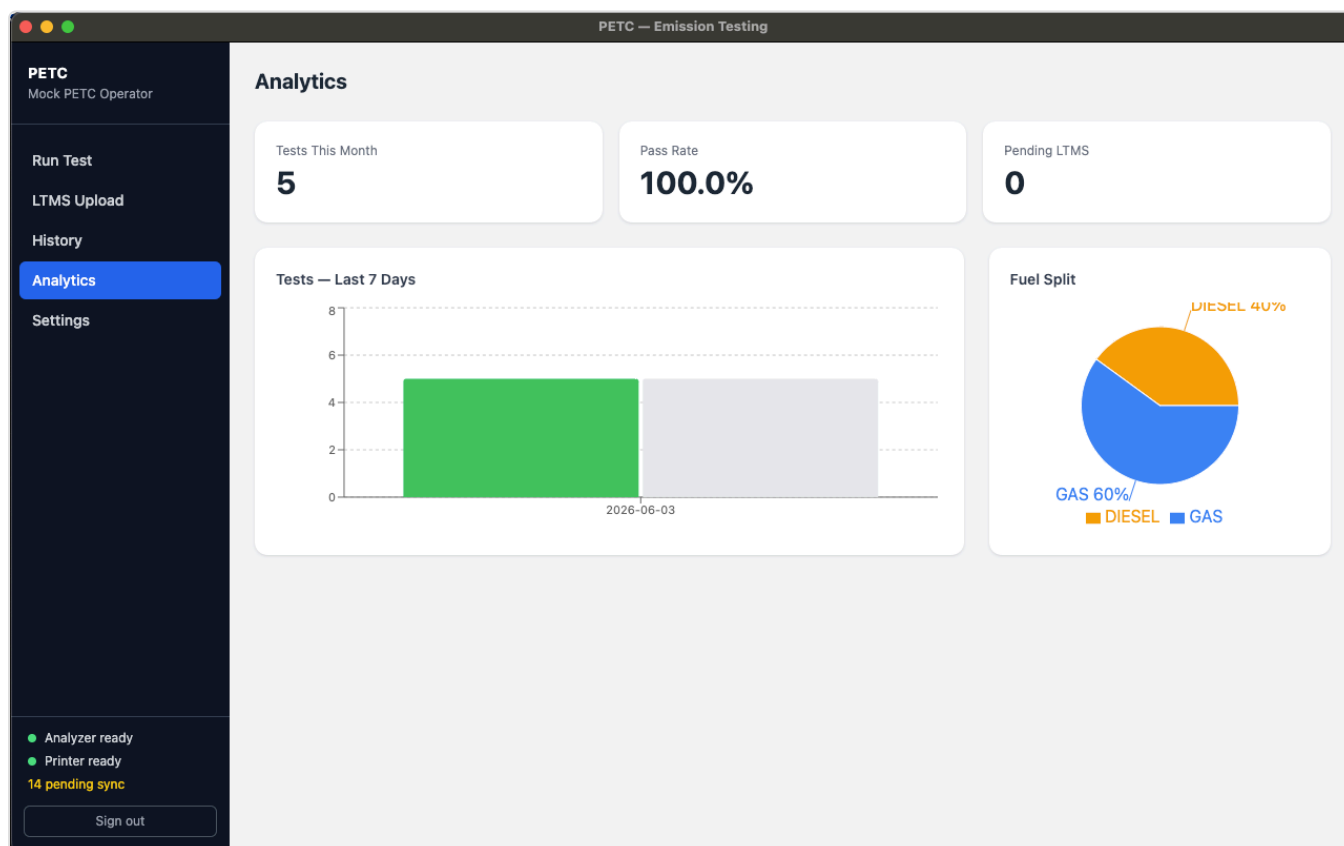


Figure 9.2 — Daily summary screen.

10. Settings

The **Settings** screen is available to supervisors and admins only.

10.1 Operators

Add, deactivate, or reset the password for operator accounts at this center. Encoders cannot be promoted to supervisor from the desktop application.

10.2 Analyzer

Choose the analyzer adapter (Fofen gas, Fofen diesel, or other supported brands), the COM port, and the baud rate. Click **Test Connection** to verify before saving.

10.3 Camera

Pick the webcam from the dropdown of detected video devices, then click **Preview**.

10.4 Printer

Pick the receipt printer and click **Print Test Page**.

10.5 Cloud

Read-only view of the Center ID, the masked center API key, and the most recent sync timestamp. To change the API key, contact Digiflash support.

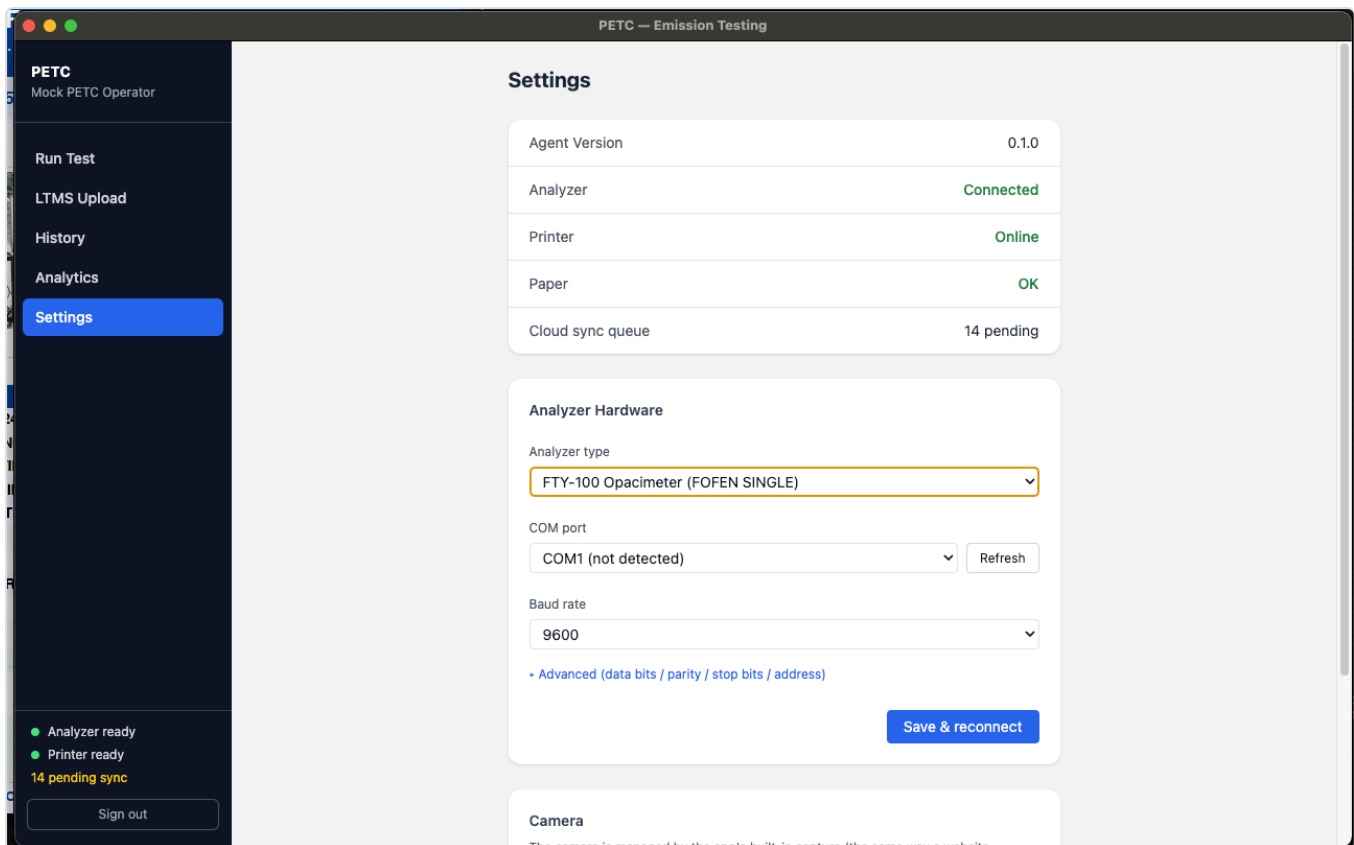


Figure 10 — Settings landing page.

11. Troubleshooting

The fixes below cover the most common day-to-day issues. Anything not listed here should be escalated to Digiflash support (Section 12).

11.1 Analyzer status shows Red

1. Check the USB-to-Serial cable between the analyzer and the PC.
2. Open **Settings** → **Analyzer** and confirm the COM port. If the port number has changed (Windows occasionally re-assigns COM numbers), update and save.
3. Power-cycle the analyzer and click **Test Connection** again.

11.2 Webcam not detected

1. Re-plug the webcam into a different USB port.
2. Open Windows **Device Manager** and confirm the webcam appears under **Cameras**. If it shows a warning icon, install or update the driver.
3. Restart Digiflash.

11.3 Printer offline or paper-out

1. Replace the receipt roll if paper-out.

2. Power-cycle the printer.
3. Open **Settings** → **Printer**, click **Print Test Page**.
4. If the test page does not print, check the USB cable.

11.4 LTMS submission rejected

The rejection banner shows the reason text returned by LTMS or IRDS. Common reasons:

Reason text contains	Fix
"Vehicle not found"	Re-verify the plate number and re-submit; if still unfound, capture the issue and contact the vehicle owner to confirm the LTMS record.
"Owner mismatch"	Re-check owner details against the LTMS lookup.
"Certification expired" / "Technician not authorised"	Verify the technician's TESDA and DOTr/LTO certification numbers on Step 4 of the wizard; if they are correct, the certification has likely expired and must be renewed.
Anything else	Read the reason verbatim, escalate to the center supervisor; the cloud and desktop both retain the full LTMS error text in the History row for support to inspect.

*The desktop does **not** generate "Duplicate submission" errors for re-attempts of the same test — the cloud is idempotent on `(center, testId)`, so re-submitting the same test returns the existing submission identifier rather than an error.*

11.5 A test in History stays in **WAITING_FOR_LTMS**

A test in **WAITING_FOR_LTMS** has reached the Digiflash cloud and the cloud is waiting for LTMS to respond. The desktop polls the cloud every 30 seconds, so a row will normally transition to **ACCEPTED** or **REJECTED** within a few minutes once LTMS replies.

1. Wait. Most rows clear within minutes.
2. If a row stays in **WAITING_FOR_LTMS** for **more than 24 hours**, check the cloud status pill on the home screen and contact Digiflash support — this generally indicates an LTMS-side outage rather than a center-side problem.
3. Do not delete or re-submit the test; the cloud will resolve it automatically when LTMS recovers.

11.6 Cloud status stays Yellow for a long time

Yellow means the desktop is saving tests locally but the Digiflash cloud is not reachable. This is normal during a temporary internet outage; tests will upload automatically when connectivity is restored. If the status stays yellow for more than one business day:

1. Check the center's internet connection (open a browser and load any website).
2. Open **Settings** → **Cloud** and verify the last sync timestamp.
3. If both internet and timestamp look normal, contact Digiflash support.

11.7 Application will not start

1. Reboot the PC.
2. If the issue persists, run the **Digiflash Repair** entry from the Start menu, which restarts the embedded services.
3. If the application still does not start, contact Digiflash support and quote the error message.

12. Support

Channel	Detail
Provider	Digiflash (DOTr-accredited IT Provider)
Product	Digiflash
Email	<i>to be provided</i>
Phone / Hotline	<i>to be provided</i>
Support hours	Monday – Saturday, 08:00 – 18:00 PHT
Emergency escalation	<i>to be provided</i>

Note for final submission: Fill in the support contact rows before submitting this manual to DOTr.

13. Glossary

Term	Meaning
CEC	Certificate of Emission Compliance. The printed certificate issued to the vehicle owner after a passing test.
Center API Key	A per-center credential that identifies the desktop application to the Digiflash cloud.
Encoder	A PETC operator who runs tests but cannot manage other operators.
IRDS	Stradcom Integrated Records and Document System.
LTMS	Land Transportation Management System, operated by LTO.
MV File Number	The Motor Vehicle file number assigned by LTO to each registered vehicle.
Outbox	The local queue of tests waiting to be uploaded to the Digiflash cloud.
Pending Upload	Test status after capture and before successful LTMS submission.
PETC	Private Emission Testing Center.

Supervisor	A PETC operator with rights to manage other operators and configure hardware.
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End of Client Application Program Manual – Digiflash PETC Data Submission Client.